

Planning Document Review of Team Kilo

1. Game Functions

- All the functions for **Matchmaking, Ranking, and Leaderboard** are well-planned and systematically structured for implementation.
 - However, here are a few suggestions for improvement:
 - **findPlayer()**: Instead of looping continuously, implement a **priority queue** where players are sorted by **ELO**. This makes potential matches more efficient.
 - **setCheckersPositions()**: Instead of iterating over the entire list, use **binary search ($O(\log n)$)** to find the player's old position and insert them into the correct spot. Also, your approach (placing tied players at the bottom of the group) is good, but consider breaking ties using the **number of games played**.
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2. Phases

- The effort to **reduce matchmaking time** by introducing **AI bots** in Phase 3 is a smart strategy, ensuring players are not stuck in long queues.
 - However, there is an inconsistency in the ranking system:
 - Phase 2 plans for **three ranks (Bronze, Silver, Gold)**.
 - The "**Getting a Rank**" section later mentions **five ranks (Bronze, Silver, Gold, Platinum, Diamond)**.
 - **This contradiction should be resolved**—decide on a final structure and ensure uniformity across all sections.
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3. Explanation of Groups Portion of Class Diagram

- Your current approach (**stopping search as soon as a valid player is found**) works well, but instead, **consider a short wait time** to allow for better matches.
 - Your plan for sorting when a player clicks the **leaderboard button** is good. **If real-time updates are needed, use lazy sorting (only sort if changes occur)**.
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4. Feedback

- The planning document **lacks sufficient detail about timelines and progress**. This makes it difficult to determine if proper deadlines exist or are being met.
- Overall, **the planning and details meet expectations**, but refinements in ranking consistency and implementation efficiency could enhance the project further.